

IEOR E4707, Spring 2026: Homework 3

February 28, 2026

This homework is due on **March 13, 2026**. You should work **individually and independently** and understand everything you write. You can apply directly a result stated in the lecture notes unless you are asked to prove it. You must show **all your work** in order to receive points. Answers without appropriate justification will receive no point. **Please refer to a Coursework announcement regarding the rule of late submission.**

Throughout this exercise a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; \{\mathcal{F}_t^W\}_{t \geq 0})$ is given in which is defined a one-dimensional Brownian motion $W = \{W_t\}_{t \geq 0}$. Moreover, $T > 0$ is fixed and given.

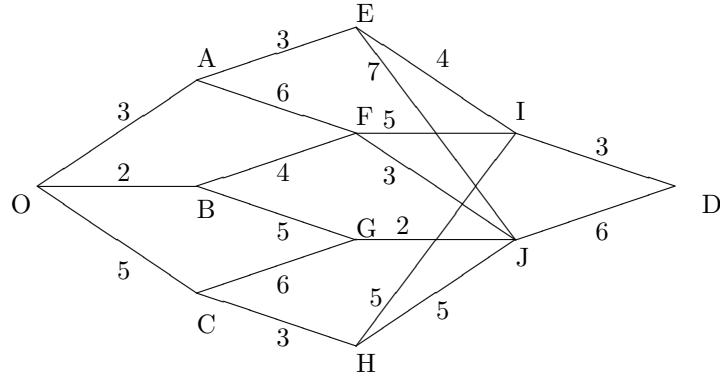
Problem 1. The following is a map where nodes (with capital letters) represent towns and the numbers shown are the distances between two directly connected towns. A traveller, starting from $t = 0$ at Town O, can choose to go to one of Towns A,B,C at $t = 1$, and then one of Towns E,F,G,H at $t = 2$, and one of Towns I, J at $t = 3$, and finally reaches Town D at $t = 4$. The traveller would like to find the shortest path from Town O to Town D along with the corresponding shortest distance he will need to travel.

a) Define the value function

$V(t, x) :=$ the shortest distance from Town x to Town D starting at time t

where $t = 0, 1, 2, 3, 4$ and x is a possible town associated with t (e.g., when $t = 1$, x is either A or B or C). Derive a relationship between $V(t, x)$ and $V(t+1, y)$ analogous to Bellman's principle of optimality (Theorem 2.1 in Lecture 9).

- b) Hence, starting from the trivial fact that $V(4, D) = 0$, derive all the values of $V(t, x)$ recursively in a backward manner including $V(0, O)$.
- c) Find the shortest path from O to D.
- d) If, half way through, the traveller finds himself deviating from the original optimal path obtained in c), how can he find the shortest path? Explain.



Problem 2. Write down the HJB equation for the following deterministic optimal control problem

$$\begin{aligned} &\text{Maximize} && J(u) = -x_T \\ &\text{s.t.} && \begin{cases} \dot{x}_t = u_t x_t \\ x_0 = 1 \\ u_t \in [-1, 1]. \end{cases} \end{aligned}$$

Problem 3. a) Write down the HJB equation for the following problem

$$\begin{aligned} &\text{Maximize} && J(u) = E \left[-\int_0^T (u_t^2 + 1) dt + \log \text{ch } x_T \right] \\ &\text{s.t.} && \begin{cases} dx_t = 2u_t dt + \sqrt{2} dW_t \\ x_0 = 0 \\ u_t \in [-1, 1] \end{cases} \end{aligned}$$

where $\text{ch } x = \frac{1}{2} (e^x + e^{-x})$.

b) Show that $V(s, y) = \log \text{ch } y$ solves the HJB equation, and hence derive the optimal feedback control based on the verification theorem.

Problem 4. (This question is necessary in completing the example in Lecture 10, Section 2.) Prove that if

$$\ln r + 2 - r < 0$$

then the following ODE

$$\dot{\theta}(t) = \frac{\theta(t)^2}{r + \theta(t)}, \quad t \in [0, 1]; \quad \theta(1) = -1$$

admits a unique solution θ satisfying $r + \theta(t) < 0 \quad \forall t \in [0, 1]$.

Problem 5. (This question is necessary in completely solving Merton's problem in a Black-Scholes market, Section 2 of Lecture 12.) Solve the ODE

$$\begin{aligned} &\dot{\theta}(t) + \gamma \left[r + \frac{\sigma^{-2} B^2}{2(1-\gamma)} \right] \theta(t) + (1-\gamma) e^{-\frac{\rho t}{1-\gamma}} \theta(t)^{\frac{\gamma}{\gamma-1}} = 0, \\ &\theta(T) = e^{-\rho T} \end{aligned}$$

and show that the solution $\theta(t) \geq 0 \ \forall t \in [0, T]$. Hence express the optimal consumption and portfolio selection strategies completely in terms of the original model parameters.

Problem 6. Derive the wealth equation (1) for the multi-asset market introduced in Lecture 12, Section 1.

Problem 7. Consider the following portfolio selection problem in a Black–Scholes market:

$$\begin{aligned} \text{Maximize} \quad & J(\pi) = \mathbb{E}[\log(x_T)], \\ \text{subject to} \quad & dx_t = (rx_t + B\pi_t)dt + \sigma\pi_t dW_t. \end{aligned}$$

Find the optimal portfolio and the value function in closed forms.